

141.  $4\text{K}_2\text{Cr}_2\text{O}_7 \xrightarrow{\text{heat}} 4\text{K}_2\text{CrO}_4 + 3\text{O}_2 + \text{X}$ . In the above reaction X is

- (a)  $\text{CrO}_3$
- (b)  $\text{Cr}_2\text{O}_7$
- (c)  $\text{Cr}_2\text{O}_3$
- (d)  $\text{CrO}_5$

**Answer: C — (c)  $\text{Cr}_2\text{O}_3$**

**Chemical reason:** Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives  $\text{Cr}_2\text{O}_3$ . The choice is consistent with the expected oxidation state and stability of the product.

142. Mond's process is used for

- (a) Ni
- (b) Al
- (c) Fe
- (d) Cu

**Answer: A — (a) Ni**

**Chemical reason:** The Mond process purifies nickel by forming volatile  $\text{Ni}(\text{CO})_4$  at low temperature and then decomposing the carbonyl at higher temperature to deposit pure Ni.

143. Stainless steel is an alloy of

- (a) Copper
- (b) Nickel and chromium
- (c) Manganese
- (d) Zinc

**Answer: B — (b) Nickel and chromium**

**Chemical reason:** Stainless steel is principally iron alloyed with chromium and commonly nickel. Chromium provides passivation; nickel improves corrosion resistance and mechanical properties.

144. Percentage of silver in German silver is

- (a) 0%
- (b) 1%
- (c) 5%
- (d) None of these

**Answer: D — (d) None of these**

**Chemical reason:** German silver contains copper, zinc, and nickel but no elemental silver. Therefore its percentage of silver is 0%.

145. Which process of purification is represented by the following scheme?

$\text{Ti}_{\text{impure}} + 2\text{I}_2 \xrightarrow{250^\circ\text{C}} \text{TiI}_4 \xrightarrow{1400^\circ\text{C}} \text{Ti}_{\text{pure}} + \text{I}_2$

- (a) Cupellation
- (b) Poling
- (c) Electrolytic refining
- (d) Zone refining
- (e) Van-Arkel process

**Answer: E — (e) Van-Arkel process**

**Chemical reason:** The correct choice is Van-Arkel process. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.



**146. Which of the following sulphides when heated strongly in air gives the corresponding metal**

- (a)  $\text{Cu}_2\text{S}$
- (b)  $\text{CuS}$
- (c)  $\text{Fe}_2\text{S}_3$
- (d)  $\text{FeS}$
- (e)  $\text{HgS}$

**Answer: E — (e)  $\text{HgS}$**

**Chemical reason:** Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives  $\text{HgS}$ . The choice is consistent with the expected oxidation state and stability of the product.

**147. Guignet's green is known as**

- (a)  $\text{Cr}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
- (b)  $\text{FeO}_3 \cdot 2\text{H}_2\text{O}$
- (c)  $\text{Cu}_2\text{O}_3$
- (d)  $\text{FeCO}_3 \cdot \text{Cr}_2\text{O}_3$
- (e)  $\text{FeO} \cdot \text{Cr}_2\text{O}_3$

**Answer: A — (a)  $\text{Cr}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$**

**Chemical reason:** This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to  $\text{Cr}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ . The choice is fixed by the known composition of that compound/mineral.

**148. Vanadium (III) oxide is a strong**

- (a) Drying agent
- (b) Oxidising agent
- (c) Reducing agent
- (d) Wetting agent
- (e) Precipitating agent

**Answer: C — (c) Reducing agent**

**Chemical reason:**  $\text{V}_2\text{O}_3$  contains vanadium in the +3 oxidation state, which is relatively low for vanadium and can be oxidized to higher states. It therefore behaves as a reducing agent.

**149. Stainless steel does not rust because**

- (a) Chromium and nickel combine with iron
- (b) Chromium forms an oxide layer and protects iron from rusting
- (c) Nickel present in it, does not rust
- (d) Iron forms a hard chemical compound with chromium present in it.

**Answer: B — (b) Chromium forms an oxide layer and protects iron from rusting**

**Chemical reason:** Chromium in stainless steel forms a thin, adherent  $\text{Cr}_2\text{O}_3$  passive film. This barrier blocks oxygen/water from continuously attacking the underlying iron, giving corrosion resistance.



**150. The main product obtained when a solution of sodium carbonate reacts with mercuric chloride is**

- (a)  $\text{Hg}(\text{OH})_2$
- (b)  $\text{HgCO}_3 \cdot \text{HgO}$
- (c)  $\text{HgCO}_3$
- (d)  $\text{HgCO}_3 \cdot \text{Hg}(\text{OH})_2$

**Answer: B — (b)  $\text{HgCO}_3 \cdot \text{HgO}$**

**Chemical reason:** Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives  $\text{HgCO}_3 \cdot \text{HgO}$ . The choice is consistent with the expected oxidation state and stability of the product.

**151. Which of the following has diamagnetic character**

- (a)  $[\text{NiCl}_4]^{2-}$
- (b)  $[\text{CoF}_6]^{3-}$
- (c)  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- (d)  $[\text{Ni}(\text{CN})_4]^{2-}$

**Answer: D — (d)  $[\text{Ni}(\text{CN})_4]^{2-}$**

**Chemical reason:** Magnetic behaviour depends on the number of unpaired d electrons.

Writing the metal-ion d configuration for the species in  $[\text{Ni}(\text{CN})_4]^{2-}$  gives the required number of unpaired electrons, so it has the stated magnetic behaviour.

**152. The solubility of silver bromide in hypo solution due to the formation of**

- (a)  $[\text{Ag}(\text{S}_2\text{O}_3)]^{-3}$
- (b)  $\text{Ag}_2\text{SO}_3$
- (c)  $[\text{Ag}(\text{S}_2\text{O}_3)]^{-}$
- (d)  $\text{Ag}_2\text{S}_2\text{O}_3$

**Answer: A — (a)  $[\text{Ag}(\text{S}_2\text{O}_3)]^{-3}$**

**Chemical reason:**  $\text{AgBr}$  dissolves in thiosulfate because  $\text{Ag}^+$  is converted into the soluble complex  $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ . Removing  $\text{Ag}^+$  from solution shifts the  $\text{AgBr}$  dissolution equilibrium forward.

**153. Brass is an alloy of**

- (a) Zn and Sn
- (b) Zn and Cu
- (c) Cu, Zn and Sn
- (d) Cu and Sn

**Answer: B — (b) Zn and Cu**

**Chemical reason:** Brass is an alloy of copper and zinc. Tin with copper gives bronze, not brass.

**154. Iodine is formed when KI reacts with a solution of**

- (a)  $\text{CuSO}_4$
- (b)  $(\text{NH}_4)_2\text{SO}_4$
- (c)  $\text{ZnSO}_4$
- (d)  $\text{FeSO}_4$

**Answer: A — (a)  $\text{CuSO}_4$**

**Chemical reason:** Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives  $\text{CuSO}_4$ . The choice is consistent with the expected oxidation state and stability of the product.



155. Rust is

- (a)  $\text{FeO} + \text{Fe}(\text{OH})_2$
- (b)  $\text{Fe}_2\text{O}_3$
- (c)  $\text{Fe}_2\text{O}_3 + \text{Fe}(\text{OH})_2$
- (d)  $\text{Fe}_2\text{O}_3$  and  $\text{Fe}(\text{OH})_3$

**Answer: C — (c)  $\text{Fe}_2\text{O}_3 + \text{Fe}(\text{OH})_2$**

**Chemical reason:** The correct choice is  $\text{Fe}_2\text{O}_3 + \text{Fe}(\text{OH})_2$ . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

156.  $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$  ion is

- (a) Colourless and diamagnetic
- (b) Coloured and octahedral
- (c) Colourless and paramagnetic
- (d) Coloured and paramagnetic

**Answer: A — (a) Colourless and diamagnetic**

**Chemical reason:** The correct choice is Colourless and diamagnetic. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

157. Which of the following is called white vitriol

- (a)  $\text{ZnCl}_2$
- (b)  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
- (c)  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
- (d)  $\text{Al}_2(\text{SO}_4)_3$

**Answer: C — (c)  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$**

**Chemical reason:** This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ . The choice is fixed by the known composition of that compound/mineral.

158.  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$  shows isomorphism

- (a)  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
- (b)  $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$
- (c)  $\text{CaSO}_4 \cdot 5\text{H}_2\text{O}$
- (d)  $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$

**Answer: A — (a)  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$**

**Chemical reason:** Isomorphous substances crystallize in the same structural type because their ions have similar charge and size. The composition represented by  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$  satisfies that structural relationship.

159. Which pair of compound is expected to show similar colour in aqueous medium

- (a)  $\text{FeCl}_2$  and  $\text{CuCl}_2$
- (b)  $\text{VOCl}_2$  and  $\text{CuCl}_2$
- (c)  $\text{VOCl}_2$  and  $\text{FeCl}_2$
- (d)  $\text{FeCl}_2$  and  $\text{MnCl}_2$

**Answer: B — (b)  $\text{VOCl}_2$  and  $\text{CuCl}_2$**

**Chemical reason:** The colour follows from the electronic structure and coordination





environment of the transition-metal ion. The species in  $\text{VOCl}_2$  and  $\text{CuCl}_2$  has the characteristic oxidation state/ligand field that produces the observed colour.

160. Which of the following dissolves in hot conc.

NaOH solution

- (a) Fe
- (b) Zn
- (c) Sn
- (d) Ag

**Answer: B — (b) Zn**

**Chemical reason:** Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result Zn is consistent with that behaviour.

